Mingen Li

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EDUCATION

University of California, Los Angeles Expected Jun 2028
M.S. in Computer Science
- Track: Artificial Intelligence
- Relevant Coursework: CS 180: Algorithms and Complexity, CS 163: Deep Learning for Computer Vision.

University of California, Berkeley Graduated May 2026
B.A. in Data Science
- Concentration: Applied Mathematics and Modeling | Technical GPA (CS, DS, Math & Stats): **3.967 / 4.0**
- Relevant Coursework: CS 61B: Data Structures, CS 188: Artificial Intelligence, CS 189: Machine Learning.

INTERNSHIP EXPERIENCE

Software Engineer, AI Systems – **Trial Data** Jun 2026 – Aug 2026
- Developed an end-to-end clinical-trial insights agent, integrating a React interface, Express APIs, and SQLite-backed data retrieval to analyze patient records, clinical test results, and study compliance.
- Built modular clinical analysis functions and task-routing logic to execute multi-step plans generated by an LLM planner.
- Built schema-aware retrieval and LLM-based validation modules to fetch task-specific clinical records and flag unsupported findings before returning results.
- Built automated evaluation scripts benchmarking the agent against a non-agentic baseline on 50 cases, increasing task success from 62% to **88%**, and reducing unsupported conclusions from 18% to **6%**.

Software Engineer, AI Systems – **Intellikgraph** Aug 2025 – Dec 2025
- Developed a maritime knowledge-graph pipeline linking vessel movements and maritime events to identify potentially suspicious vessel behavior and interactions.
- Modeled and ingested vessel trajectory and maritime event data into AllegroGraph, linking related entities and events for graph-based analysis.
- Designed and implemented SPARQL-based verification for maritime anomalies, checking LLM-generated assessments against graph evidence, achieving 92.5% (vs. 76.3% baseline) across 80 synthetic cases.

RESEARCH EXPERIENCE

Research Intern – **NSF REU (Computer Vision)** May 2025 – Aug 2025
Advisor: Dr. Yu Tian, Assistant Professor, University of Central Florida
- Developed a training-free video anomaly detection pipeline integrating BLIP-2 for frame captioning and LLaVA-OneVision for visual reasoning to analyze surveillance footage.
- Implemented hierarchical frame- and segment-level reasoning with FAISS-backed dynamic memory to maintain contextual information for LLM-based anomaly assessment.
- Achieved **81.13%** AUC on UCF-Crime vs. 78.32% for zero-shot SOTA LAVAD under identical settings.

SKILLS

Languages: **Python**, Java, C++, JavaScript, SQL, SPARQL.
Frameworks & Tools: React, Node.js, Express.js, SQLite, Git, AllegroGraph.
AI/ML: PyTorch, Transformers, LLM Agents, RAG, LLM Evaluation